📐 MATHEMATICS FOR MACHINE LEARNING

CampusX · Data Science Mentorship Program 2023 · Video 9 of 23

SESSION 45
Hypothesis Testing Part 1

Null & Alternative Hypotheses · Type I & II Errors · P-Values · Z-Tests

Hypothesis Testing Framework · Null vs Alternative Hypothesis · Type I & Type II Errors · Significance Level (Alpha) · Decision Rules: Critical Value vs P-Value · One-Tailed vs Two-Tailed Tests · One-Sample Z-Test · Two-Sample Z-Test (A/B Testing)

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Instructor: Nitish Singh (CampusX)

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01 · Hypothesis Testing Framework

Hypothesis Testing is a formal statistical method used to make decisions about a population based on sample data. It is widely used in machine learning for feature selection, model evaluation, and A/B testing.

02 · Null vs Alternative Hypothesis

Every hypothesis test involves two competing statements:

- **Null Hypothesis (H_0):** The status quo, assuming no effect, no change, or no difference. We assume H_0 is true unless proven otherwise.
- **Alternative Hypothesis (H_a or H_1):** The claim we want to support, suggesting there is a significant effect or difference.

03 · Type I & Type II Errors

DECISION / REALITY	H_0 IS TRUE	H_0 IS FALSE
Reject H_0	Type I Error (α) (False Positive)	Correct Decision ($1-\beta$) (Power of the Test)
Fail to Reject H_0	Correct Decision (No Error)	Type II Error (β) (False Negative)

04 · Significance Level (Alpha)

The **significance level (\$≣lpha\$)** is the maximum probability of committing a Type I error. It is set by the researcher before the test begins (usually \$≣lpha = 0.05\$ or \$0.01\$).

05 · Decision Rules: Critical Value vs P-Value

We make decisions using one of two equivalent approaches:

- **Critical Value Approach:** Reject H_0 if the computed test statistic is in the critical region.
- **P-value Approach:** Reject H_0 if the **p-value $\le ≡lpha$**. The p-value is the probability of obtaining a test statistic at least as extreme as the observed value, assuming H_0 is true.

06 · One-Tailed vs Two-Tailed Tests

- **One-Tailed:** H_a specifies a direction (e.g., $\mu > \mu_0$ or $\mu < \mu_0$). All critical area is on one side of the distribution.
- **Two-Tailed:** H_a does not specify a direction ($\mu eq \mu_0$). The critical area is split equally between both tails.

07 · One-Sample Z-Test

Used to compare a sample mean to a known population mean when population variance σ^2 is known.

Z-STATISTIC FORMULA

$$Z = (\bar{X} - \mu_0) / (\sigma / \sqrt{n})$$

08 · Two-Sample Z-Test (A/B Testing)

Used in A/B testing to compare the conversion rates or means of two independent groups (control and treatment):

TWO-SAMPLE Z-STATISTIC

$$Z = (\hat{p}_1 - \hat{p}_2) / \sqrt{[p_{pool}(1 - p_{pool})(1/n_1 + 1/n_2)]}$$

09 · Quick Reference Card

🚀 Hypothesis Testing part 1 Cheat Sheet

- Type I Error (α): Rejecting H_0 when it is actually true (False Positive).
- Type II Error (β): Failing to reject H_0 when it is false (False Negative).
- Rule: If **p-value $\le ≣lpha$**, reject H_0 (statistically significant).
- One-Sample Z-Test: Used only when population standard deviation is known.